



# Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in $pp$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Phenomenology of the Standard Model of Elementary Particles

Master's Degree in Physics - Elementary Particles and

Astroparticles

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# Table of Contents

## 1 Introduction

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



# Probing the EW Sector and BSM Physics with $W^\pm Zjj$

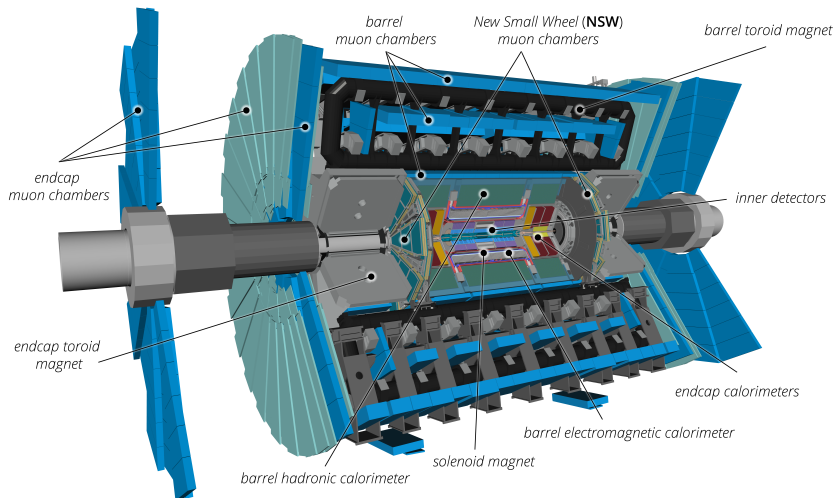
## 1 Introduction

- **Dataset:**  $140 \text{ fb}^{-1}$  of  $pp$  collisions at  $\sqrt{s} = 13 \text{ TeV}$  (2015–2018).
- **Target:** Fully leptonic decays of  $W^\pm Zjj$  production.
- **Measurements:** Integrated & differential fiducial cross-sections ( $\sigma_{\text{EW}}$  and  $\sigma_{\text{strong}}$ ).
- **Goal:** Isolate Vector Boson Scattering (VBS).
- **New Physics:** EFT limits on Anomalous Quartic Gauge Couplings (aQGC).



# The ATLAS Detector

## 1 Introduction





# Table of Contents

## 2 Theoretical Framework

- ▶ Introduction
- ▶ **Theoretical Framework**
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



# Gauge Boson Self-Interactions

## 2 Theoretical Framework

- The Electroweak (EW) sector of the Standard Model is based on the **non-abelian** gauge symmetry  $SU(2)_L \times U(1)_Y$ .
- Unlike QED, the generators of  $SU(2)_L$  do not commute. As a consequence, the physical vector bosons ( $W^\pm, Z, \gamma$ ) self-interact.
- This structure directly dictates the existence of **Triple** (TGC) and **Quartic** (QGC) Gauge Couplings:

$$\mathcal{L}_{\text{TGC}} = \frac{1}{2} g \epsilon_{abc} W_b^\mu W_c^\nu (\partial_\mu W_{a\nu} - \partial_\nu W_{a\mu}) \quad \mathcal{L}_{\text{QGC}} = \frac{1}{4} g^2 \epsilon_{abc} \epsilon_{ars} W_b^\mu W_c^\nu W_{r\mu} W_{s\nu}$$

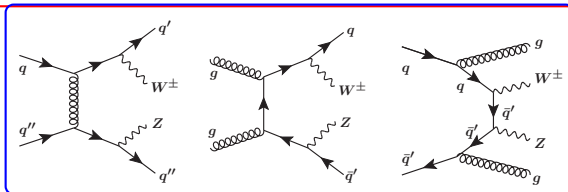
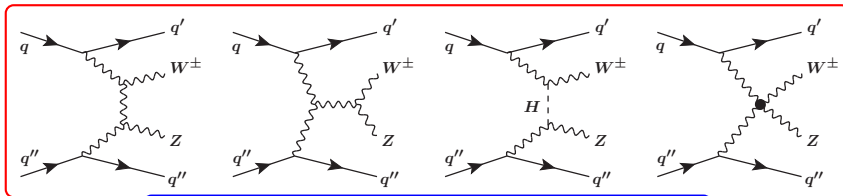
Probe these vertices to test the SM rigidity and search for anomalous couplings (aQGC) arising from New Physics.



# Experimental Signature: Vector Boson Scattering

## 2 Theoretical Framework

**Vector Boson Scattering (VBS)**  $VV \rightarrow VV$  with  $V = W/Z/\gamma$ , is a key process with which to probe the  $SU(2)_L \times U(1)_Y$  gauge symmetry of the **electroweak** (EW) theory that determines the **self-couplings** of vector bosons. To study these couplings, we exploit the purely electroweak production of  $W^\pm Zjj$ .





# A little look at the Effective Field Theory (EFT)

## 2 Theoretical Framework

The  $W^\pm Zjj$ -EW production can be sensitive to effects beyond the SM affecting the quartic interactions of weak bosons. The measured  $W^\pm Zjj$  events are therefore used to search for anomalous quartic gauge couplings (aQGC) using an EFT framework. In this model, the Lagrangian of the standard model is extended into an efficient Lagrangian by adding higher-order operators and their respective Wilson coefficients as follows:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i^{(6)}}{\Lambda^2} O_i^{(6)} + \sum_j \frac{f_j^{(8)}}{\Lambda^4} O_j^{(8)} + \dots$$

Where:

- $O_{i,j}^{(6),(8)}$ : are dimension-6 and dimension-8 operators;
- $f_i^{(6)}$  e  $f_j^{(8)}$ : dimensionless couplings
- $\Lambda$ : is the energy scale of the new processes





# A little look at the Effective Field Theory (EFT)

## 2 Theoretical Framework

The squared scattering amplitude of the effective field theory prediction for  $W^\pm Z jj$  production can be written as:

$$\left| A_{\text{SM}} + \sum_i c_i A_i \right|^2 = |A_{\text{SM}}|^2 + \sum_i c_i 2 \text{Re}(A_{\text{SM}}^* A_i) + \sum_i c_i^2 |A_i|^2 + \sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$$

- $c_i = f_j^{(8)} / \Lambda^4$ ,  $A_{\text{SM}}$  is the SM scattering amplitude;
- $2 \text{Re}(A_{\text{SM}}^* A_i)$  is the amplitude of the interference term between the SM and the dimension-8 operators;
- $\sum_i c_i^2 |A_i|^2$  is the pure dimension-8 contribution, and  $\sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$  is the amplitude of interferences between two dimension-8 operators, called cross terms.



# Table of Contents

## 3 Fiducial Phase Space and MC Simulations

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ **Fiducial Phase Space and MC Simulations**
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



# Fiducial Phase Space for Leptons

## 3 Fiducial Phase Space and MC Simulations

The  $W^\pm Z jj$  cross-sections are measured in a fiducial phase space. Leptons produced in the decay of a hadron, a  $\tau$ -lepton, or their descendants are not considered in the definition of the fiducial phase space and are treated as background.

| Category          | Kinematic Variable                                                | Requirement        |
|-------------------|-------------------------------------------------------------------|--------------------|
| Z leptons         | Transverse momentum ( $p_T^{\ell_Z}$ )                            | $> 15 \text{ GeV}$ |
| W lepton          | Transverse momentum ( $p_T^{\ell_W}$ )                            | $> 20 \text{ GeV}$ |
| All leptons       | Pseudorapidity ( $ \eta_\ell $ )                                  | $< 2.5$            |
| Z boson           | Invariant mass window ( $ m_{\ell^+\ell^-} - m_Z^{\text{PDG}} $ ) | $< 10 \text{ GeV}$ |
| W boson           | Transverse mass ( $m_T^W$ )                                       | $> 30 \text{ GeV}$ |
| Angular isolation | Distance between Z leptons ( $\Delta R(\ell_{Z1}, \ell_{Z2})$ )   | $> 0.2$            |
| Angular isolation | Distance between W and Z leptons ( $\Delta R(\ell_W, \ell_Z)$ )   | $> 0.3$            |



# Fiducial Phase Space for Jets

## 3 Fiducial Phase Space and MC Simulations

In addition, at least two jets are required, reconstructed using the anti- $k_t$  algorithm with radius parameter  $R = 0.4$ . Hard processes with a  $b$ -quark in the initial or final states, such as  $tZj$  production, are not considered as part of the  $W^\pm Zjj$ -EW events.

| Category          | Kinematic Variable                                        | Requirement            |
|-------------------|-----------------------------------------------------------|------------------------|
| All jets          | Transverse momentum ( $p_T^j$ )                           | $> 40 \text{ GeV}$     |
| All jets          | Pseudorapidity ( $ \eta_j $ )                             | $< 4.5$                |
| Angular isolation | Distance between jets and leptons ( $\Delta R(j, \ell)$ ) | $> 0.3$                |
| Tagging jets      | Opposite hemispheres ( $\eta_{j1} \cdot \eta_{j2}$ )      | $< 0$                  |
| Tagging jets      | Invariant mass ( $m_{jj}$ )                               | $> 500 \text{ GeV}$    |
| Jet Veto          |                                                           |                        |
| $b$ -quark jets   | Absence of $b$ -jets in initial/final state               | $N_{b\text{-jet}} = 0$ |



## How to eliminate leptons ambiguity?

### 3 Fiducial Phase Space and MC Simulations

- Leptons and final-state neutrons that do not originate from hadron or  $\tau$ -lepton decays, are matched to the  $W^\pm$  and  $Z$  boson decay products using an algorithmic approach, called the **resonant shape algorithm**.
- This algorithm is based on the value of an estimator expressing the product of the nominal line-shapes of the  $W$  and  $Z$  resonances:

$$P = \left| \frac{1}{m_{(\ell^+, \ell^-)}^2 - (m_Z^{\text{PDG}})^2 + i \Gamma_Z^{\text{PDG}} m_Z^{\text{PDG}}} \right|^2 \times \left| \frac{1}{m_{(\ell', \nu_{\ell'})}^2 - (m_W^{\text{PDG}})^2 + i \Gamma_W^{\text{PDG}} m_W^{\text{PDG}}} \right|^2$$



# Monte Carlo Simulation of the Signal

## 3 Fiducial Phase Space and MC Simulations

The simulation of  $W^{\pm}Zjj$  events was primarily generated using the MadGraph5\_aMC@NL02.6.5 MC event generator, interfaced with Pythia8.240. The simulation was divided into distinct components:

- **Electroweak production  $W^{\pm}Zjj$  – EW:** The actual signal. It is calculated at leading order (LO), considering purely electroweak processes (order  $\alpha_{EW}^6$ ).
- **QCD production  $W^{\pm}Zjj$  – QCD:** Considered a background for the VBS analysis. This component is calculated at next-to-leading order (NLO) in QCD.
- **Interference  $W^{\pm}Zjj$  – INT:** Quantifies the quantum interference between the EW and QCD production. It is calculated at LO and yields a positive contribution of 6% in the fiducial region, reaching up to 12% for events with three or more jets.



# Monte Carlo Simulation of the Background

## 3 Fiducial Phase Space and MC Simulations

In addition to the  $W^{\pm}Zjj$  – QCD production, there are numerous other Standard Model processes that can mimic the signal in the detector. These backgrounds fall into two categories based on how they are estimated:

- **Backgrounds with *prompt (real)* leptons:** Processes such as the production of  $ZZ$  pairs,  $WW$ , tribosons ( $VVV$ ), or top quarks associated with vector bosons ( $t\bar{t}V$ ,  $tZj$ ). Their contribution is evaluated entirely using Monte Carlo simulations.
- **Backgrounds with *fake* leptons:** Events where a hadronic jet is misidentified as a lepton by the detector. Being an instrumental artifact that is complex to simulate accurately, this background is estimated directly using data-driven techniques.



## Detector Simulation

### 3 Fiducial Phase Space and MC Simulations

All generated MC events were passed through the ATLAS detector simulation, based on Geant4, and processed using the same reconstruction software used for the data. Furthermore, the lepton and jet momentum scale and resolution, the lepton reconstruction, identification, isolation and trigger efficiencies and the pile-up jets and b-jet veto efficiencies in the simulation were corrected to match those measured in data.





# Table of Contents

## 4 Event selection & Background estimation

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ **Event selection & Background estimation**
- ▶ Measurement methodology and results
- ▶ Conclusion



# Event Selection: Physics Objects Definition

## 4 Event selection & Background estimation

### Data Quality & Trigger:

- Only data with stable beams and fully operational sub-detectors.
- Single-lepton triggers used (Combined efficiency  $\approx 99\%$  for offline-selected  $W^\pm Zjj$  events).

### Physics Objects Reconstruction:

- **Prompt Leptons**  $e, \mu, \nu$  ( $E_T^{miss}$ ): Reconstructed combining ID tracks with ECAL clusters or MS tracks. Strict isolation and impact parameter criteria are applied to reject non-prompt background.
- **Hadronic Jets**: Clustered using the *anti- $k_t$*  algorithm ( $R = 0.4$ ).
- **b-tagging**: Deep-learning algorithms identify displaced vertices from b-hadron decays (crucial for top-quark veto).



# Physics Objects: Leptons

## 4 Event selection & Background estimation

| Object    | Selection Criteria                                                                                                                                                                                                                                                                |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Muons     | <i>Medium</i> Identification (Inner Detector + Muon Spectrometer)<br>$p_T > 15 \text{ GeV}$ , $ \eta  < 2.5$<br>Impact parameters: $ d_0/\sigma_{d_0}  < 3.0$ , $ z_0 \sin \theta  < 0.5 \text{ mm}$<br>Isolation efficiency: 90 – 99% (depending on $p_T$ )                      |
| Electrons | <i>Medium</i> Likelihood Identification<br>$p_T > 15 \text{ GeV}$ , $ \eta  < 1.37$ or $1.52 <  \eta  < 2.47$ (crack region veto)<br>Impact parameters: $ d_0/\sigma_{d_0}  < 5.0$ , $ z_0 \sin \theta  < 0.5 \text{ mm}$<br>Isolation efficiency: 89 – 99% (depending on $p_T$ ) |

*Note: Impact parameter cuts are crucial to select **prompt** leptons and reject non-prompt background from heavy-flavor decays.*



## Physics Objects: Jets & $E_T^{\text{miss}}$

4 Event selection & Background estimation

| Object                             |        | Selection Criteria                                                                                                                                                                                                                                                      |
|------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jets                               |        | Clustered via <i>anti-<math>k_t</math></i> algorithm ( $R = 0.4$ )<br>$p_T > 25 \text{ GeV}$ , $ \eta  < 4.5$<br>Pile-up suppression: <i>tight</i> ( $p_T < 60 \text{ GeV}$ , $ \eta  < 2.4$ ) and <i>fJVT loose</i> ( $p_T < 120 \text{ GeV}$ , $2.5 <  \eta  < 4.5$ ) |
| Missing<br>( $E_T^{\text{miss}}$ ) | Energy | Proxy for the undetected neutrino.<br>Computed as the negative vector sum of all calibrated <i>hard</i> objects (leptons, jets) plus a track-based <i>soft</i> term.                                                                                                    |



# Event Selection: Boson Reconstruction

## 4 Event selection & Background estimation

| Category          | Selection Criteria                                                                                                                                               |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseline          | Exactly 3 lepton candidates ( $\mu$ or $e$ )<br>Veto on a 4th <i>loose</i> lepton ( $p_T > 5$ GeV)<br>Trigger matching: at least 1 lepton with $p_T > 25/27$ GeV |
| Z Boson Candidate | Opposite-Sign Same-Flavor (OSSF) lepton pair<br>$ m_{\ell\ell} - m_Z^{\text{PDG}}  < 10$ GeV<br>(If ambiguous, the pair closest to $m_Z$ is chosen)              |
| W Boson Candidate | The remaining 3rd lepton + $E_T^{\text{miss}}$<br>Stricter requirements: <i>Tight</i> ID & Isolation, $p_T > 20$ GeV<br>Transverse mass: $m_T^W > 30$ GeV        |



## Signal Region (SR): VBS Topology

4 Event selection & Background estimation

| Category        | Selection Criteria                                                                                                                    |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Tagging Jets    | The two highest- $p_T$ jets in the event ( $p_T > 40$ GeV)<br>Must reside in opposite hemispheres ( $\eta_{j1} \cdot \eta_{j2} < 0$ ) |
| Kinematic Cuts  | $m_{jj} > 150$ GeV (rejects triboson background)<br>$m_{jj} > 500$ GeV (isolates the EWK VBS process from QCD)                        |
| Background Veto | <b>Exact b-veto</b> ( $N_{b\text{-jet}} = 0$ )<br>(strongly suppresses top-quark background)                                          |



# Background Estimation Strategy

## 4 Event selection & Background estimation

### Two Main Categories:

- **Irreducible:** True prompt leptons mimicking the signal topology.
- **Reducible (Fakes):** Non-prompt leptons or hadronic jets misidentified as prompt leptons.

### Estimation Techniques in the Simultaneous Fit:

- **Data-driven (Control Regions):** Used for dominant backgrounds. Extract normalization factors ( $\mu_{bkg}$ ) directly from data.
- **Pure Monte Carlo:** Used for rare processes. Constrained by theoretical uncertainties.
- **Matrix Method:** Fully data-driven approach for reducible fakes, bypassing MC fragmentation modelling.



# Irreducible Backgrounds

## 4 Event selection & Background estimation

### Constrained via Control Regions (CRs):

- $ZZ \rightarrow 4\ell$  (**ZZjj-QCD**): Survives via lepton acceptance loss. Constrained in the orthogonal **ZZ-CR** (requiring a 4th *loose* lepton).
- **Top-quark** ( $t\bar{t} + V, tZj$ ): Survives via b-tagging inefficiency. Constrained in the **b-CR** (inverting the b-veto,  $\geq 1$  b-jet).

### Estimated via Pure Monte Carlo:

- **Tribosons** ( $VVV$ ) & **ZZjj-EW**: Yields too low to define a dedicated CR.
- Handled as Gaussian *Nuisance Parameters* ( $\theta$ ) in the Profile Likelihood Ratio.





# Reducible Backgrounds (Fake Leptons)

## 4 Event selection & Background estimation

### Sources:

- $Z + \text{jets}, Z\gamma, t\bar{t}$  without a true 3rd prompt lepton.
- Light/heavy-flavor jets decaying semi-leptonically or photon conversions.

### The Matrix Method:

- Relies on two lepton identification criteria: *Loose* (baseline) and *Tight* (signal).
- Uses real lepton efficiency ( $\epsilon$ ) and fake rate ( $f$ ) measured in independent data samples.
- Solves a linear system to estimate the fake contribution directly in the Signal Region.



# Expected and Observed Numbers of Events

## 4 Event selection & Background estimation

|                    | SR ( $N_{\text{jet}} = 2$ ) | SR ( $N_{\text{jet}} \geq 3$ ) | $b$ -CR         | ZZ-CR             |
|--------------------|-----------------------------|--------------------------------|-----------------|-------------------|
| <b>Data</b>        | 169                         | 477                            | 666             | 210               |
| <b>Total pred.</b> | $231 \pm 12$                | $550 \pm 50$                   | $660 \pm 40$    | $205 \pm 11$      |
| $W^{\pm}Zjj$ — EW  | $65.0 \pm 3.5$              | $60 \pm 6$                     | $4.82 \pm 0.28$ | $0.725 \pm 0.014$ |
| $W^{\pm}Zjj$ — QCD | $125 \pm 9$                 | $380 \pm 50$                   | $77 \pm 18$     | $6.2 \pm 0.7$     |
| $W^{\pm}Zjj$ -INT  | $1.3 \pm 0.6$               | $5.3 \pm 2.6$                  | $0.58 \pm 0.29$ | $0.22 \pm 0.11$   |
| $t\bar{t} + V$     | $0.66 \pm 0.04$             | $20.2 \pm 0.7$                 | $289 \pm 10$    | $9.89 \pm 0.28$   |
| $tZj$              | $8.78 \pm 0.34$             | $19.7 \pm 1.2$                 | $134 \pm 4$     | $0.432 \pm 0.005$ |
| ZZ-QCD             | $9.6 \pm 0.4$               | $32.0 \pm 2.5$                 | $10.1 \pm 0.6$  | $159 \pm 9$       |
| ZZ-EW              | $2.2 \pm 0.6$               | $4.4 \pm 1.1$                  | $0.25 \pm 0.06$ | $23 \pm 6$        |
| VVV                | $0.41 \pm 0.10$             | $2.0 \pm 0.5$                  | $0.39 \pm 0.10$ | $4.1 \pm 1.1$     |
| Misid. leptons     | $18 \pm 4$                  | $28 \pm 7$                     | $150 \pm 40$    | $1.7 \pm 0.5$     |



# Table of Contents

## 5 Measurement methodology and results

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ **Measurement methodology and results**
- ▶ Conclusion



# Signal Extraction: Multivariate Analysis (BDT)

## 5 Measurement methodology and results

### The Challenge:

- $WZjj$ -EW (signal,  $\mathcal{O}(\alpha_{EW}^6)$ ) and  $WZjj$ -QCD (background,  $\mathcal{O}(\alpha_s^2 \alpha_{EW}^4)$ ) share the exact same final state.
- Cut-based analysis is sub-optimal  $\Rightarrow$  Boosted Decision Tree (TMVA).

### BDT Input Variables (15 observables):

- *Jet Kinematics*:  $m_{jj}$ ,  $\Delta\eta_{jj}$ ,  $\Delta\phi_{jj}$ , jet multiplicity.
- *Boson Kinematics*:  $p_T^W$ ,  $p_T^Z$ ,  $m_T^{WZ}$ .
- *Mixed Correlations*: Z-centrality ( $\zeta_{lep}$ ), event balance.

### BDT Output:

- **The Score**: A single output that allows you to differentiate the signal.

*Note: The BDT is also used to differential the background in the b-CR between  $t\bar{t}V$   $tZj$ .*



# The Global Profile Likelihood Fit

## 5 Measurement methodology and results

### Simultaneous Extended Binned Negative Log-Likelihood:

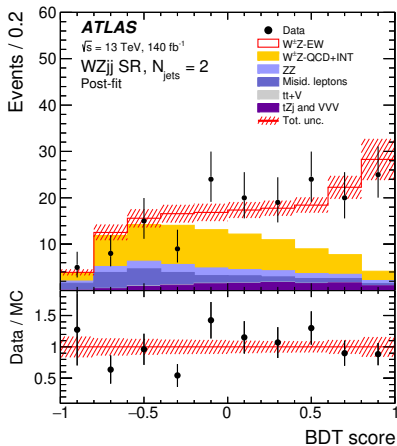
$$-\ln \mathcal{L}(\mu_{sig}, \mathbf{K}, \boldsymbol{\theta}) = \sum_r \sum_i \left( \mu^{[r,i]} - N^{[r,i]} \ln \mu^{[r,i]} \right) + \sum_j \frac{\theta_j^2}{2}$$

- **Regions ( $r$ ):** SR ( $N_{jets} = 2$ ), SR ( $N_{jets} \geq 3$ ), b-CR, ZZ-CR.
- **Observables ( $N^{[r,i]}$ ):** BDT score in SRs,  $m_{jj}$  in ZZ-CR, b-CR BDT in b-CR.
- **Parameters of Interest (POI):** Signal strengths  $\mu_{sig}$  for EWK and QCD.
- **Unconstrained Parameters ( $\mathbf{K}$ ):** Background normalizations ( $K_{t\bar{t}V}$ ,  $K_{ZZ}$ ) extracted dynamically from CRs.
- **Nuisance Parameters ( $\boldsymbol{\theta}$ ):** Theoretical and experimental systematics, constrained by the Gaussian penalty term  $\theta_j^2/2$ .

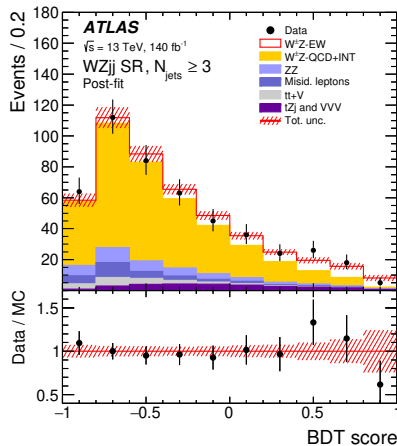


# Post-fit distributions SR ( $N_{jets} = 2$ ), SR ( $N_{jets} \geq 3$ )

5 Measurement methodology and results



(a)

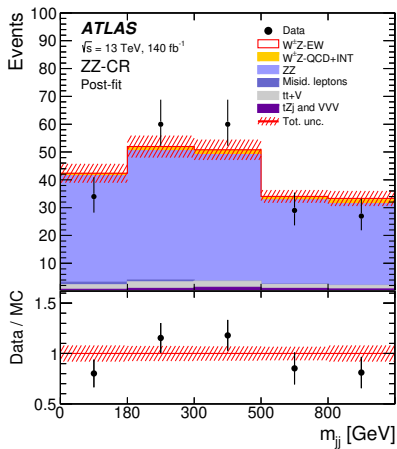


(b)

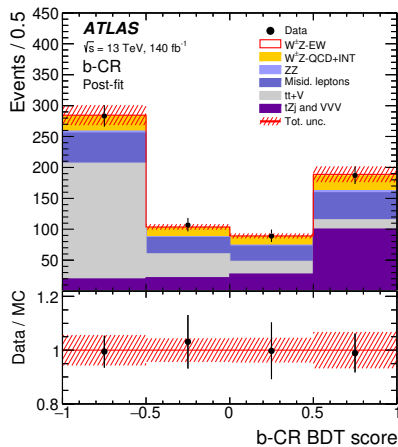


# Post-fit distributions ZZ-CR, b-CR

## 5 Measurement methodology and results



(c)



(d)



# Fiducial Cross-Section Extraction

## 5 Measurement methodology and results

The fiducial cross-sections are extracted directly from the profile-likelihood fit. The normalisation factors ( $\mu_{EW}$ ,  $\mu_{QCD}$ ) scale the theoretical MC predictions within the fiducial phase space:

$$\sigma_{WZjj-EW} = \sum_i \mu_i^{WZjj-EW} \cdot \sigma_{i,th.MC}^{WZjj-EW}$$
$$\sigma_{WZjj-strong} = \sum_i \left( \mu_i^{WZjj-QCD} \cdot \sigma_{i,th.MC}^{WZjj-QCD} + \mu_i^{WZjj-INT} \cdot \sigma_{i,th.MC}^{WZjj-INT} \right)$$

*\*Index  $i$  runs over the two SR categories:  $N_{jets} = 2$  and  $N_{jets} \geq 3$ .*

- **Interference term:**  $\mu_{INT}$  is constrained as  $\sqrt{\mu_{EW}} \cdot \sqrt{\mu_{QCD}}$ .
- **Categorization:** Splitting by  $N_{jets}$  minimizes the dependency on the MC modelling of QCD radiation.





# Results and Comparison with SM Predictions

## 5 Measurement methodology and results

### VBS (EWK Production)

$$\sigma_{\text{obs}}^{\text{EWK}} = 0.368 \pm 0.037_{\text{stat}} \pm 0.059_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{EWK}} = 0.370 \pm 0.030 \text{ fb}$$

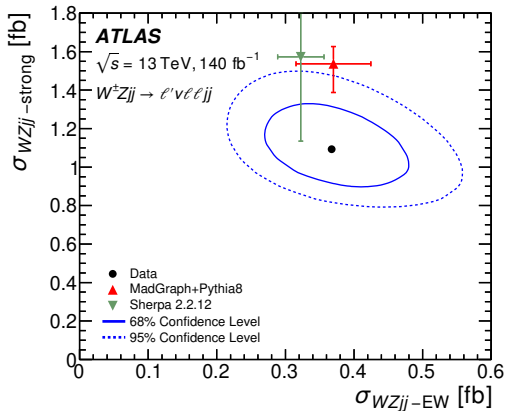
⇒ Excellent SM agreement

### QCD + Int. (Strong Production)

$$\sigma_{\text{obs}}^{\text{QCD}} = 1.093 \pm 0.066_{\text{stat}} \pm 0.131_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{QCD}} = 1.537 \pm 0.150 \text{ fb}$$

⇒ Deficit factor  $\sim 0.7$  ( $1.8\sigma$  tension)





# Table of Contents

## 6 Conclusion

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



# Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in $pp$ collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

*Thank you for listening!*

*Any questions?*